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Task NO: 01

Q NO: 01

Design a C++ program to create a class named BankAccount. The class should contain the following private data members:

- accountNumber — to store the account number
- accountHolder — to store the account holder's name
- balance — to store the account balance

Create a public member function named setData() that receives the account number, account holder's name, and balance as arguments and stores them in the private data members.

In the main() function:

1. Create an object of the BankAccount class.
2. Use the setData() function to store the following information:
 - Account Number: 101
 - Account Holder: "Ali"
 - Balance: 50000
3. Create another object and store different account information.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class BankAccount
```

```
{
```

```
private:
```

```
    int accountNumber;
```

```
    string accountHolder;
```

```
    double balance;
```

```
public:
```

```
    void setData(int a, string b, double c)
```

```
{
```

```
    accountNumber = a;
```

```
    accountHolder = b;
```

```
    balance = c;
```

```
}
```

```
    void display()
```

```
{
```

```
        cout << "Account Number: " << accountNumber << endl;

        cout << "Account Holder: " << accountHolder << endl;

        cout << "Balance: " << balance << endl;

    }

};
```

```
int main()

{

    BankAccount A1;

    A1.setData(101, "Ali", 50000);

    A1.display();

    return 0;

}
```

Q NO: 02

Design a C++ program to create a class named Student. The class should contain the following **private data members**:

- studentID — to store the student's ID
- studentName — to store the student's name
- department — to store the student's department
- semester — to store the student's current semester

Create a **public member function** named setData() that receives the student ID, student name, department, and semester as arguments and stores them in the private data members.

In the main() function:

1. Create an object of the Student class.
2. Use the setData() function to store the following information:
 - Student ID: 101
 - Student Name: "Ali"
 - Department: "Computer Science"
 - Semester: 3
3. Create another object and store different student information.
4. Use the appropriate access modifiers to ensure that the student information cannot be accessed directly outside the class.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class Student
```

```
{
```

```
private:
```

```
    int studentID;
```

```
    string studentName;
```

```
    string department;
```

```
    int semester;
```

```
public:
```

```
    void setData(int a, string b, string c, int d)
```

```
{
```

```
    studentID = a;
```

```
    studentName = b;
```

```
        department = c;  
        semester = d;  
    }
```

```
void display()  
{  
    cout << "Student ID: " << studentID << endl;  
    cout << "Student Name: " << studentName << endl;  
    cout << "Department: " << department << endl;  
    cout << "Semester: " << semester << endl;  
}  
};
```

```
int main()  
{  
    Student S1;  
  
    student1.setData(101, "Ali", "Computer Science", 3);  
    S1.display();  
  
    return 0;  
}
```